INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE FOR VEHICLE

<table>
<tr><td rowspan="2">ELECTRIC

PROPULSION</td><td>VEHICLE NAME/MODEL:

McNEILUS® VOLTERRA™ ZFL™
FRONT LOADER</td><td rowspan="2">TYPE OF REESS:
LITHIUM-ION, 700V battery storage system</td></tr>
<tr><td>VEHICLE TYPE/DESIGNATION
Sub-Category:
Heavy Heavy Duty Urban</td></tr>
<tr><td colspan="3"></td></tr>
<tr><td colspan="3">Document PN: 1695896
Document Revision: 12/2025</td></tr>
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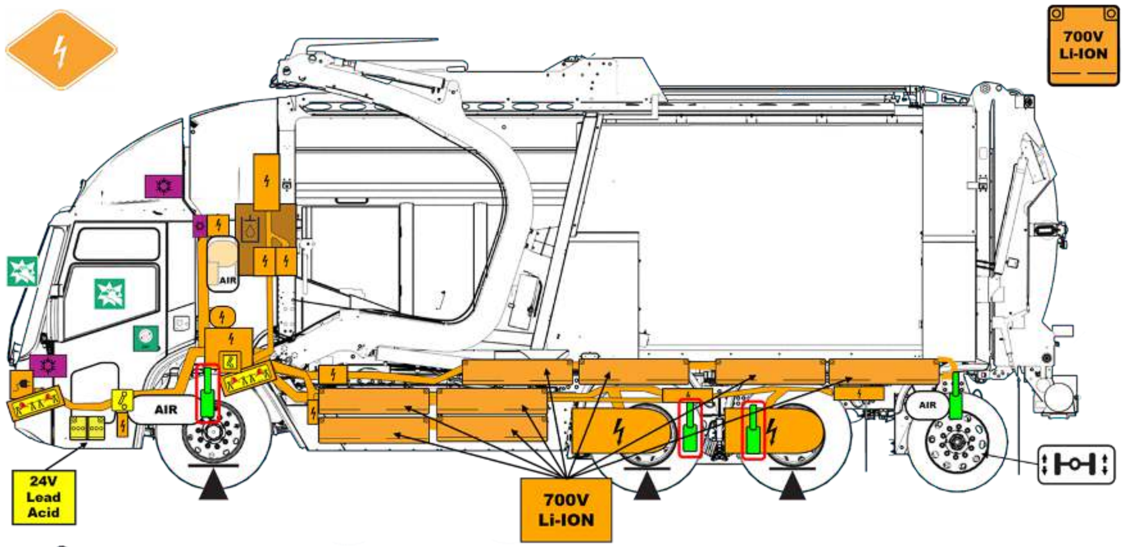
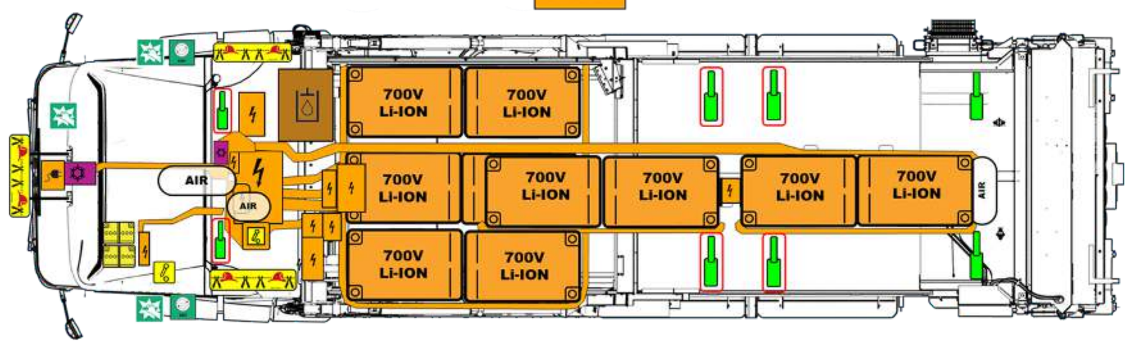
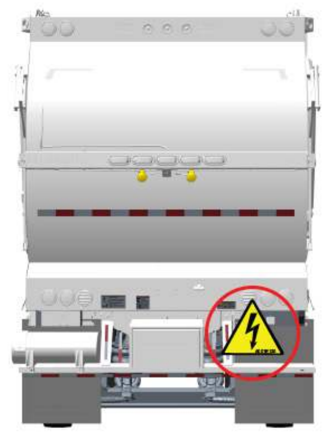
McNeilus	VEHICLE NAME/MODEL: McNeilus® Volterra™ ZFL ™ VEHICLE TYPE/DESIGNATION: Front Loader PRODUCTION YEARS: 2025 — Present	





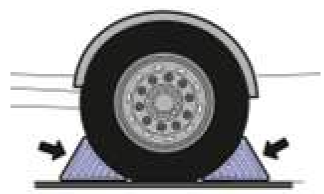
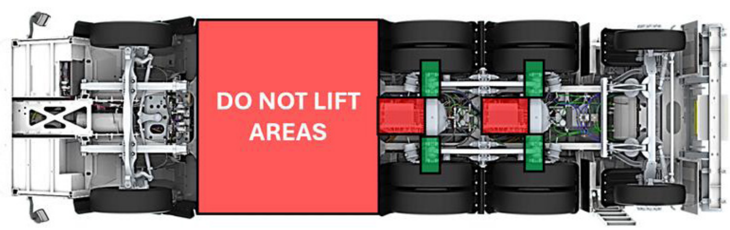
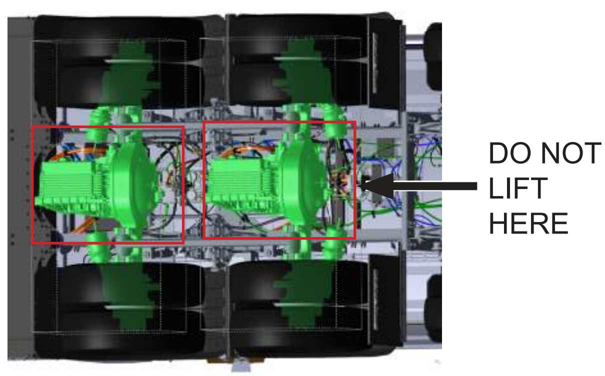
High Voltage (HV) Battery 620V nominal 700V maximum	Low Voltage (LV) 12V Lead Acid battery - 24V system	High Voltage (HV) Component	Lifting Point	Air Conditioning Component
Low Voltage (LV) Disconnect	AIR Compressed Air Tank	Gas Strut/ Preloaded Spring	High Voltage (HV) Disconnect	EXIT Entry/Exit Points
High Voltage (HV) Power Cable	Moving Tag Axle	Break Points	Hydraulic Fluid Reservoir	High Voltage (HV) Charging Inlet
High Voltage (HV) Disconnect Cut Loop		**NOTE:** These images show information for one variant of this product. The number of traction batteries, location and number of axles, and body configuration can vary depending on the product.		

1. Identification / Recognition	
Electric (700 VDC Lithium-Ion Battery)	
VIN location An Oshkosh Corporation Vehicle Identification Data Plate contains the model and Vehicle Identification Number (VIN). The last six (6) digits of the VIN is the vehicle serial number. The VIN and ID tag will follow primary driver side (LH drive will have VIN and ID tag on streetside; RH drive will have VIN and ID tag on curbside). If the vehicle is dual drive, both the plate and tag will be on the streetside driver's door.	
Front and Rear Views	 
Streetside and Curbside Views	 
Electrification Badge	VOLTERRA ZFL™

2. Immobilization / Stabilization / Lifting

⚠ **WARNING: Unintended Movement Hazard. Failure to follow shutdown procedures could result in unintended vehicle movement. Always approach the vehicle from the sides to stay out of the potential travel path and properly chock the wheels. Reduced vehicle noise may make it difficult to determine if the truck is energized. Failure to comply may result in serious personal injury or death.**

⚠ **WARNING: Lifting Hazard. Do not use high voltage battery or any high voltage components to lift or stabilize vehicle. Use only designated lifting points. Failure to comply may result in serious personal injury or death.**

Park Brake Pull out to apply brake. Push in to release brake. Applying the parking brake automatically shifts the vehicle to neutral (N).	
Wheel Chocks Properly chock wheels to prevent vehicle movement. Use only wheel chocks provided with the vehicle located inside the rear toolbox under the tailgate or on either side of the body above the rear fenders.	
Lifting/Jacking the vehicle Use jacking points identified in the illustration. The illustration shows view from underside of the vehicle. Legend 🟩 Lift/Jack location 🟥 DO NOT LIFT/JACK	 
Righting the vehicle if it is on its side The body is very rigid. Chains should be used to upright the vehicle. Ensure lifting devices do not contact high voltage cables or batteries.	

3. Disable direct hazards / Safety regulations

WARNING: Electrical Shock Hazard. Always assume the vehicle's high voltage system is powered ON! Cutting, touching, crushing, or tampering with the vehicle's high voltage components can result in serious injury or death. Always use appropriate tools and use appropriate personal protective equipment (PPE). Failure to comply may result in serious personal injury or death.

WARNING: Electrical Shock Hazard. De-Energizing the high voltage (HV) system does not dissipate voltage inside the battery pack! HV battery packs remain energized. High voltage cables between battery packs and HV power distribution unit remain energized. Making contact with the high voltage battery pack internals or battery cables may result in serious personal injury or death.

WARNING: Electrical Shock Hazard. Electric vehicles damaged in a crash may have compromised safety systems and may present a potential electrical shock hazard. Use caution and use proper protective equipment including high voltage safety gloves and boots. Remove all metallic jewelry. Isolate the high voltage system as directed. Failure to comply may result in serious personal injury or death.

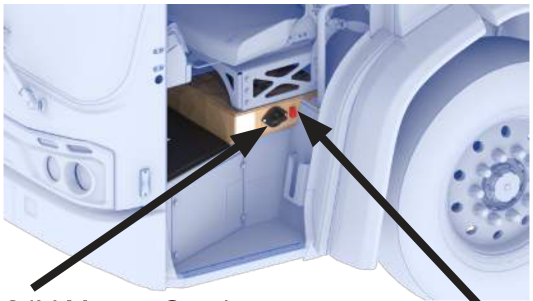
WARNING: Electrical Shock Hazard. Service and handling of high voltage components is to be restricted to qualified personnel only. Failure to follow this instruction and those applicable to your region may result in serious personal injury or death.

3.1 Approaching a Vehicle

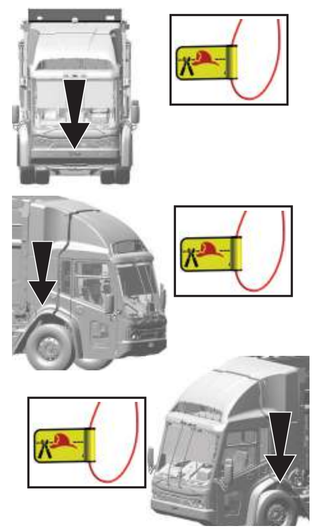
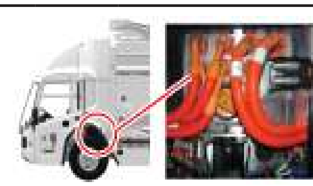
Remove all metallic conducting items such as jewelry before approaching the vehicle. Always chock the wheels to restrict vehicle movement.

3.2 Disabling Procedure

1. Engage Park Brake.  	Pull out to apply brake. Push in to release brake. • Applying the parking brake automatically shifts the vehicle to neutral (N).	
2. Turn off ignition.	 	

3. Deactivate the Low Battery Deactivate the **Low Battery** via the 24V Master Service Disconnect Switch   	1. Confirm chassis parking brake is applied and wheels are chocked. 2. Confirm ignition is OFF. 3. Remove key from vehicle ignition and maintain in personal possession. 4. Shut off toggle switch next to the 24V Master Service Disconnect Switch. 5. Wait 10 seconds. 6. Turn 24V Master Service Disconnect Switch to OFF. 7. Perform your company’s lockout/tagout procedures.

3.3 Alternative Methods for Disabling the High Voltage System

Option 1: High Voltage Disconnect (cut loop)	• Locate the most accessible cut loop from the Electrification Diagram and cut a section out so the ends cannot touch.	
Option 2: Deactivate High Voltage	• Turn off the High Voltage Manual Disconnect switch, located on the street side of the high voltage power distribution unit (HVPDU).	 

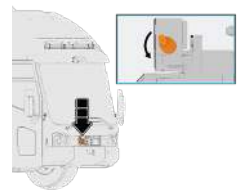
3.4 Deactivation Method if Vehicle is Charging

If vehicle is actively charging,

- Activate E-STOP at the charging port location if available on electric vehicle supply equipment (EVSE, which includes charging station, cable, connectors, etc.).
- Wait for green light to turn off.
- Unplug vehicle from charger.
- Unlock cab.
- Turn the 24V Master Service Disconnect Switch located below the street side seat to the OFF position.

If charge inlet port is locked,

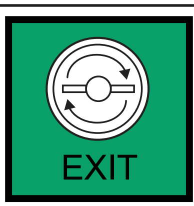
- Activate E-STOP at the charging port location if available on electric vehicle supply equipment (EVSE, which includes charging station, cable, connectors, etc.).
- Turn the 24V Master Service Disconnect Switch located below the street side (left hand side) seat to the OFF position.
- Turn the High Voltage Manual Disconnect switch to the OFF position (it is located on the street side of the high voltage power distribution unit [HVPDU]).
- Actuate the mechanical lock on the charge inlet.



4. Access to the vehicle's occupants

⚠ **WARNING: Pinch Point Hazard. Vehicle doors are heavy. Use caution when opening and avoid pinch points. Failure to comply may result in serious personal injury or death.**

⚠ **WARNING: Fall Hazard. Always maintain three (3) points of contact (such as one foot and two hands or one hand and two feet) when getting in or out of the vehicle. Use step and hand holds provided. Failure to comply may result in serious personal injury or death.**



Two door exits.

To open the doors from outside, pull the door handle out toward you and pull door to open.

To open the doors from inside, pull the door handle toward you and push out on the door to open.

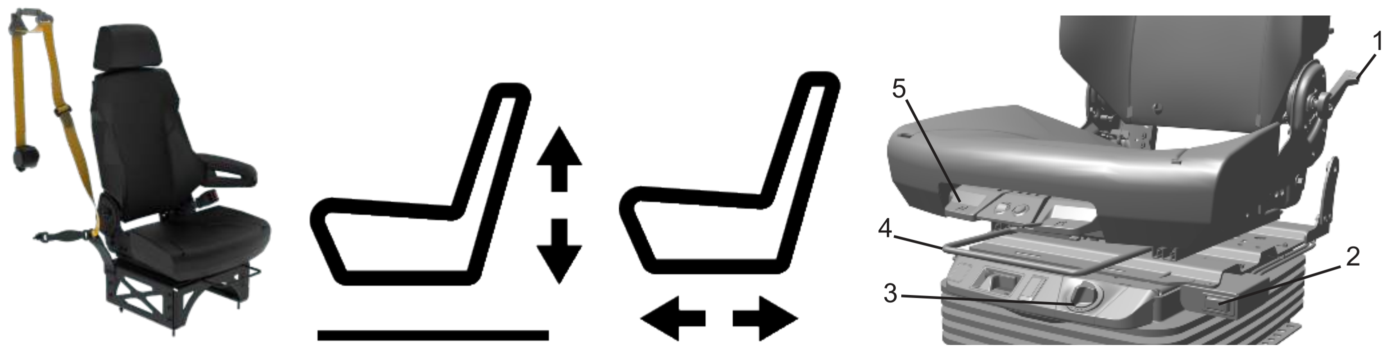
To open door outside cab

To open door from inside cab

Break these windows to obtain access.

Windshield (1) is safety glass per FMVSS and ANSI. Remove windshield first if accessible. Remove windows (2) as needed.

Seat Adjustment and Seat Restraints



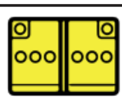
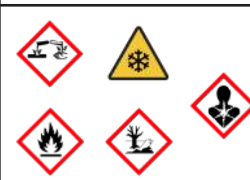
1. **Back Rest Adjust Control** – Control lever adjusts back rest position.
2. **Height Adjust Air Control Valve** – Valve raises or lowers seat position.
3. **Shock Adjust Control** – Adjusts the shock absorption amount that the air ride seat takes.
4. **Fore/Aft Slide Control** – Lift bar allows seat to slide fore/aft for desired position.
5. **Cushion Extension Control** – Control allows extension of the seat cushion.

Steering Column Adjustment

Pull tilt/telescopic lever toward you to extend steering column

Push tilt/telescopic level away from you to lower steering column

5. Stored Energy / Liquids / Solids

Component	Type	Capacity	Color	Hazards Associated
 HV battery	700 V LI-ON	450-598 kWh	Colorless	
 LV battery	24 V Lead Acid	96 Ah	Colorless	
 Refrigerant	R134A	R134a - 3.5 lbs (500 PSI)	Blue-Green	
Hydraulic Oil	ISO 32	50 Gallons (2750 PSI)	Dark Brown	
Ethylene Glycol Coolant	50-50	22 Gallons (35 PSI)	Pink-Red	

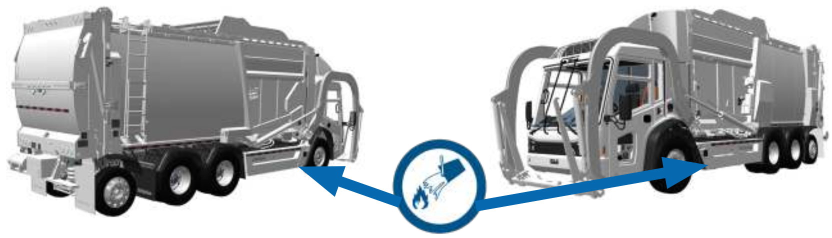
6. In case of Fire

WARNING: Inhalation Hazard. Fires in crash-damaged electric vehicles may emit toxic or combustible gasses. Wear personal protective equipment and self-contained breathing apparatus (SCBA) when working in close proximity or in a confined area. Ventilate the vehicle interior by opening the vehicle windows or doors. Ventilate the working area. Furthermore, take appropriate measures to protect civilians downwind from the incident. Failure to comply may result in serious personal injury or death.

WARNING: Explosion Hazard. Lithium-Ion batteries can explode during fire. Use caution and wear proper PPE. Failure to comply may result in serious personal injury or death.

WARNING: Fire Hazard. Electric vehicles with damaged high voltage batteries require special handling precautions. Inspect the vehicle carefully for leaking battery fluids, sparks, flames, and gurgling or bubbling sounds. Contact emergency services immediately if any of these problems are observed. Failure to comply may result in a vehicle fire and serious personal injury or death.

6.1 For Vehicle Fires	
Extinguishing Fire	Use Wet Foam when other materials are involved. Use water to extinguish LI-ion fires.
Hazardous to Human Health	May cause an allergic skin reaction. Do not breathe dust, fumes, gas, mist, vapors, or spray. Breather valves may emit large flames because of thermal runaway.
Explosion Hazard	Explosive gas could accumulate – ventilate the working area. Move the vehicle outside buildings after extinguishing fire. Damaged LI-ion batteries could reignite; keep vehicle away from combustible materials.
Corrosives	Causes skin burns and eye damage.

High Voltage (705 V) 	CAT III (1000 V) rated PPE required for exposed high voltage (HV) parts.
Battery Fire 	• May take 24 hours for battery pack to cool. • Remove sources of ignition. • Cool burned mass with water to cool battery cells. • Check LI-ion battery pack fires with thermal infrared camera (TIC or IR gun)
6.2 For Battery System Fires Flood the forward-located battery structure with water to cool structure.	
6.3 For Refuse Body Fires • Move vehicle to a safe area. • Eject the trash from the body if possible. • Extinguish the fire with water and/or a fire extinguisher.	

7. In case of Submersion

DANGER: Battery Failure Present Hazard. Visible bubbling or fizzing from a submerged or damaged electric vehicle may indicate high-voltage battery failure or an active electrical discharge. Do not touch, move, or attempt extraction of the vehicle while bubbling or fizzing is present. Maintain a safe distance until the reaction has stopped and the area is deemed safe. Failure to comply will result in serious personal injury or death.

WARNING: Electric Shock Hazard. High-voltage components and wiring may be energized and present a shock hazard. Always treat the high-voltage system as fully charged. Use appropriate PPE, including high-voltage safety gloves and boots, and remove all metallic objects such as jewelry before approaching or handling the vehicle. Failure to comply may result in serious personal injury or death.

WARNING: Electric Shock Hazard. A flooded vehicle that is plugged into a charging station can pose an electrocution hazard from the vehicle to the charging station. Always make sure that the vehicle is not plugged into a charging station before attempting to service. Failure to comply may result in a vehicle fire and serious personal injury or death.

WARNING: Fire Hazard. Flooded battery can still have stored energy and catch on fire.

WARNING: Electric Shock Hazard. Vehicles that have been submerged have a higher risk for a high voltage electrical battery fire and should be handled with more caution. The damage level of a submerged vehicle may not be visible. Submersion in water can damage the electrical components. It is recommended that fire fighters be present and ready in case of fire.

- Avoid any contact with the high voltage cables and electric components. If possible, remove the vehicle from the water and continue with the disable procedure for this vehicle (see Chapter 3. Disable direct hazards / Safety regulations for more information). Do not remove High Voltage (HV) Disconnects from HV batteries or modify any HV components or HV Cables.

8. Towing / Transportation / Storage

WARNING: Electric Shock Hazard. Do not tow the vehicle with wheels on the ground without disengaging the axles as this may cause the vehicle to generate electricity and can cause potential damage. Failure to comply may result in serious personal injury or death.

WARNING: Unintended Movement Hazard. Always approach the vehicle from the sides to stay out of the potential travel path. It may be difficult to determine if the vehicle is running due to lack of engine noise. Failure to comply may result in serious personal injury or death.

WARNING: Fire Hazard. Be alert. There is a potential for delayed fire with damaged lithium-ion batteries. Measure temperature with thermal camera before and after transportation. Failure to comply may result in a vehicle fire and serious personal injury or death.

CAUTION: Do not move or tow the vehicle with wheels on the ground if center display Maintenance screen does not state “ALLOWED TO TOW” as this may cause the vehicle to generate electricity and can cause potential damage. If shifting is non-functional, manually apply air pressure to shift exhaust tube of both rear axles. Failure to comply may result in damage to the vehicle.

8.1 Towing / Transportation Procedure

Front wheel lift is the preferred method of towing the vehicle. See illustrations below.

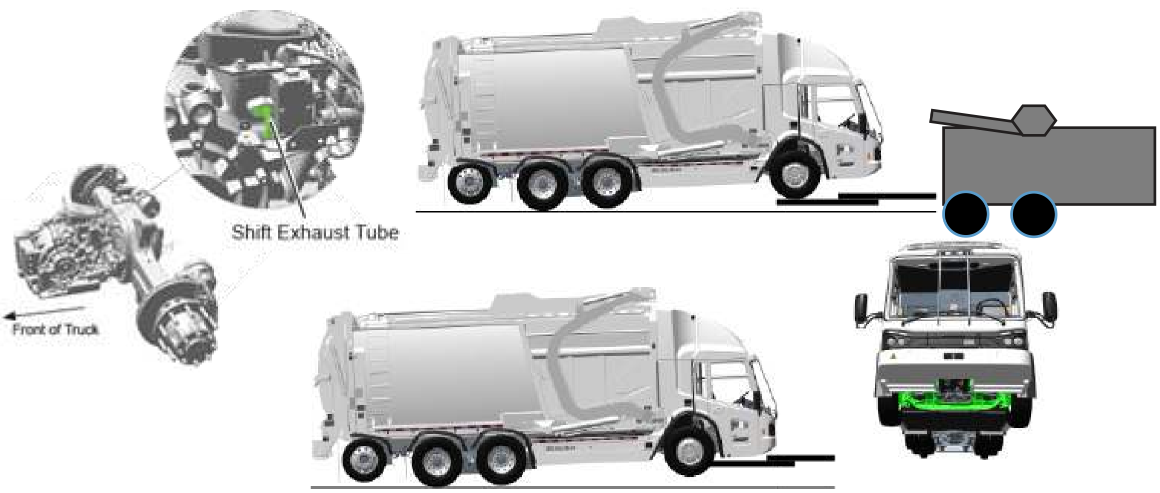
Towing Vehicle with Empty Body

- Front wheel lift with shift selector in neutral (N).
- Front axle lift with shift selector in neutral (N) — lift from front axle spring mounts.

Towing Vehicle a Loaded Body

- Front wheel lift with shift selector in neutral (N).

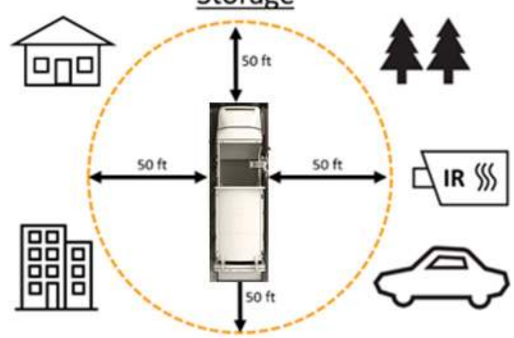
NOTE: If selector is not functional, manually apply air pressure (90-120 PSI) into the shift exhaust tube of both rear axles. Confirm center display MAINTENANCE screen states “ALLOWED TO TOW”.



8.2 Storage Procedure

- Make sure the damaged vehicle is kept in an open area 50 feet away from structures, combustibles, and other vehicles. Do **NOT** in an enclosed building.
- If you see leaking fluids, sparks, smoke, flames, increased temperature, gurgling, popping, or hissing noises from the high voltage battery compartment, ventilate the area and call 911.
- Procedures:
  - Place the vehicle in Neutral
  - Set the parking brake
  - Turn off the vehicle
  - Activate hazard lights
  - Chock the vehicle wheels

Storage



9. Important Additional Information

The McNeilus® Volterra™ comes in multiple variants. The rescue sheet image shows information for one variant of this product. The number of traction batteries, location and number of axles, and body configuration can vary depending on the variant of the product.

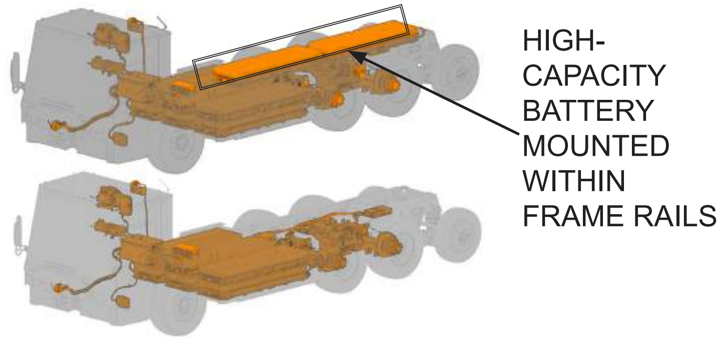
This includes two (2x) different battery configurations (medium and high-capacity) and two (2x) unique wheelbase lengths (185” and 208”) can vary from vehicle to vehicle.

BATTERY SYSTEM CONFIGURATION

The standard medium capacity battery system contains only batteries between the front and tandem axles.

The high string capacity system contains batteries mounted in between frame rails.

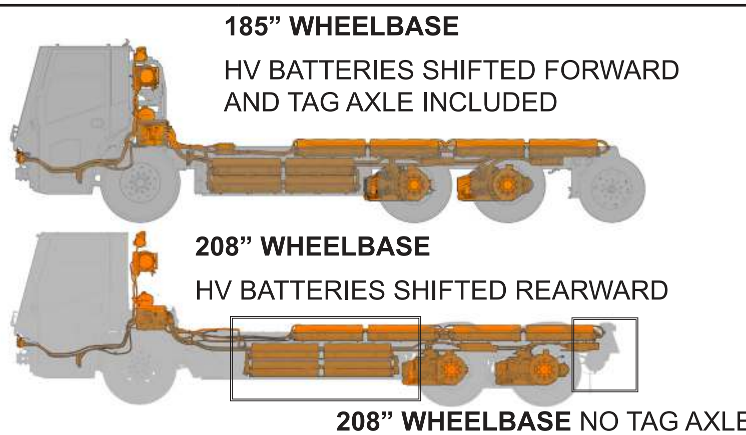
Emergency response procedures are common between both variants' battery variants.



WHEELBASE CONFIGURATIONS

The vehicle comes in two wheelbase lengths: 185" and 208" wheelbase. The 185" wheelbase includes a tag axle. Whereas the 208" wheelbase shifts batteries and tandem e-axles rearward and removes the tag axle.

Disabling procedures and emergency response procedures are common between both variants.



10. Explanation of Pictograms Used

Use water to extinguish fire

Use ABC powder to extinguish fire

Environmental Hazard

Flammable

Explosive

Corrosive

Hazardous to Human Health

Acute Toxicity

Warning: High Voltage

General Warning

Warning: Low Temperature